SMART INDIA HACKATHON 2026

Smart Study Assistant (Dual-Mode Learning Engine)

- **Problem Statement ID – SIH260207**
- **Problem Statement Title - Student Innovation-Smart education, a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably.**
- **Theme - Smart Education**
- **PS Category - Software**
- **Team ID - SIH26-A0H-T115**
- **Team Name (Registered on portal) - JARVIX**

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Smart Study Assistant (Dual-Mode Learning Engine)

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•Proposed Solution:

- A personalized learning platform that diagnoses weak topics and provides targeted micro-lessons and practice questions.
- Works on smartphones and PCs, plus basic 2G feature phones via zero-internet USSD codes (*123#) and SMS.

•How It Solves the Problem:

- Replaces uniform classroom teaching with custom, self-paced learning paths for every student.
- Eliminates digital divide barriers so rural students without internet can still practice and learn.

•Key Innovations:

- **Root-Cause Detection:** Isolates basic formula gaps before teaching advanced concepts.
- **Zero-Internet USSD Mode:** Interactive quizzes on simple button phones without mobile data.
- **Offline Sync & Voice:** Offline study caching and read-aloud tools for full accessibility

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TECHNICAL APPROACH

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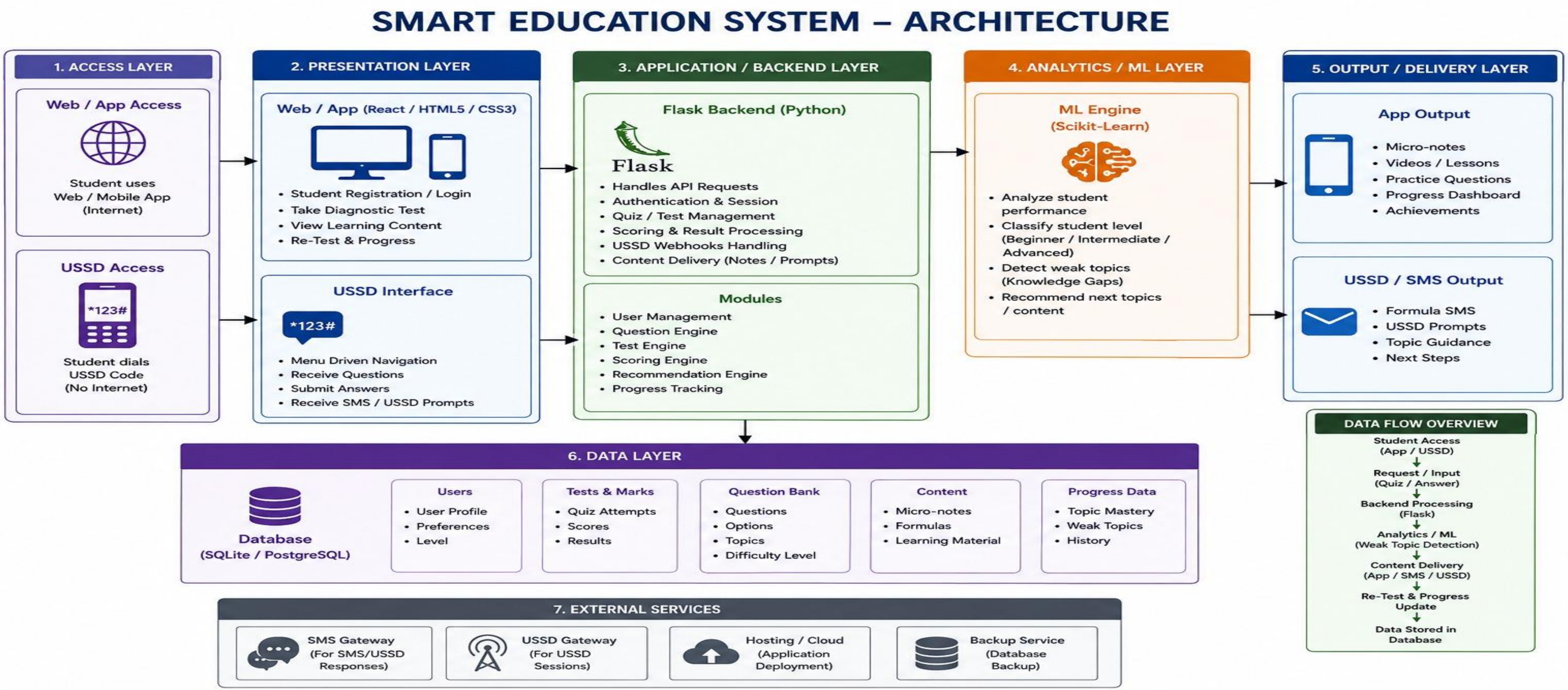

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FEASIBILITY AND VIABILITY

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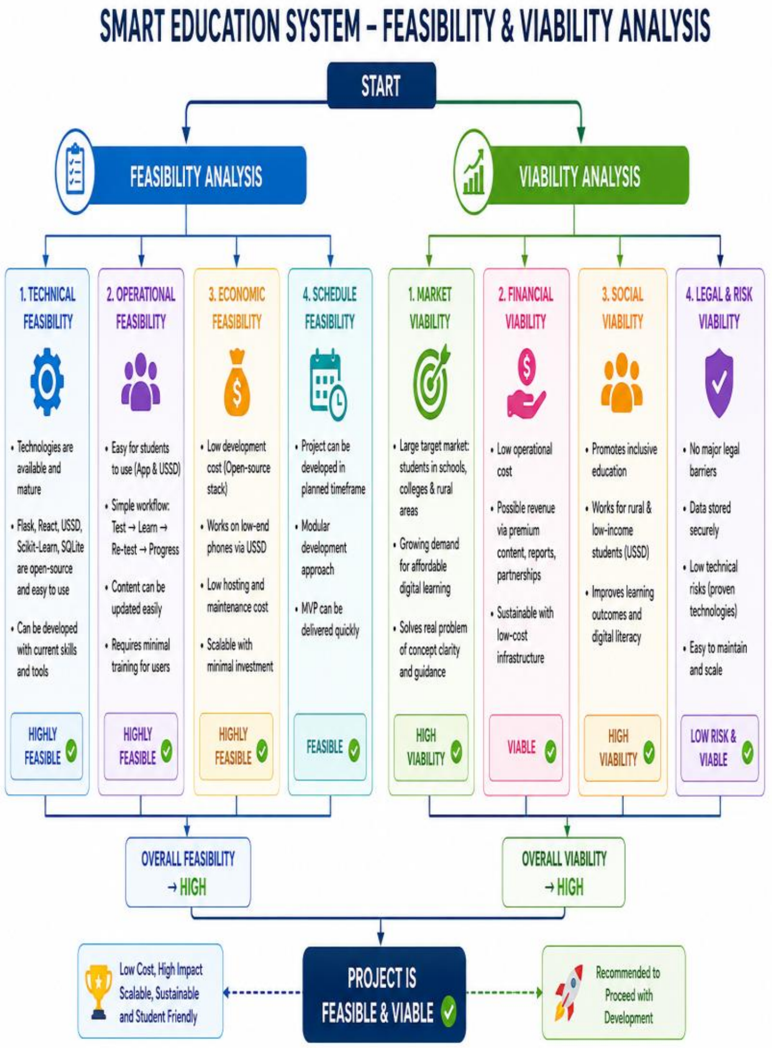
Feasibility & Viability:

- **Low Cost & High Scalability:** Built entirely on free, open-source technologies (Python, Flask, SQLite) with no expensive hardware or server requirements.
- **Universal Compatibility:** Works seamlessly across modern web browsers, low-end smartphones, and basic 2G button phones.



Key Challenges & Solutions:

- **Zero / Unstable Internet:**
  - *Solution:* Offline-first caching for smartphones plus a USSD (*123#) and SMS fallback for basic feature phones.
- **New Learner Profiling (Cold Start):**
  - *Solution:* A mandatory 5-question baseline quiz during signup instantly establishes the student's skill level.
- **USSD Screen & Character Limits:**
  - *Solution:* Questions, micro-notes, and formulas are modularized into concise, 160-character single-screen prompts.
- **Data Security & Privacy:**
  - *Solution:* End-to-end user authentication and encrypted database records protect student progress and personal data.

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IMPACT AND BENEFITS

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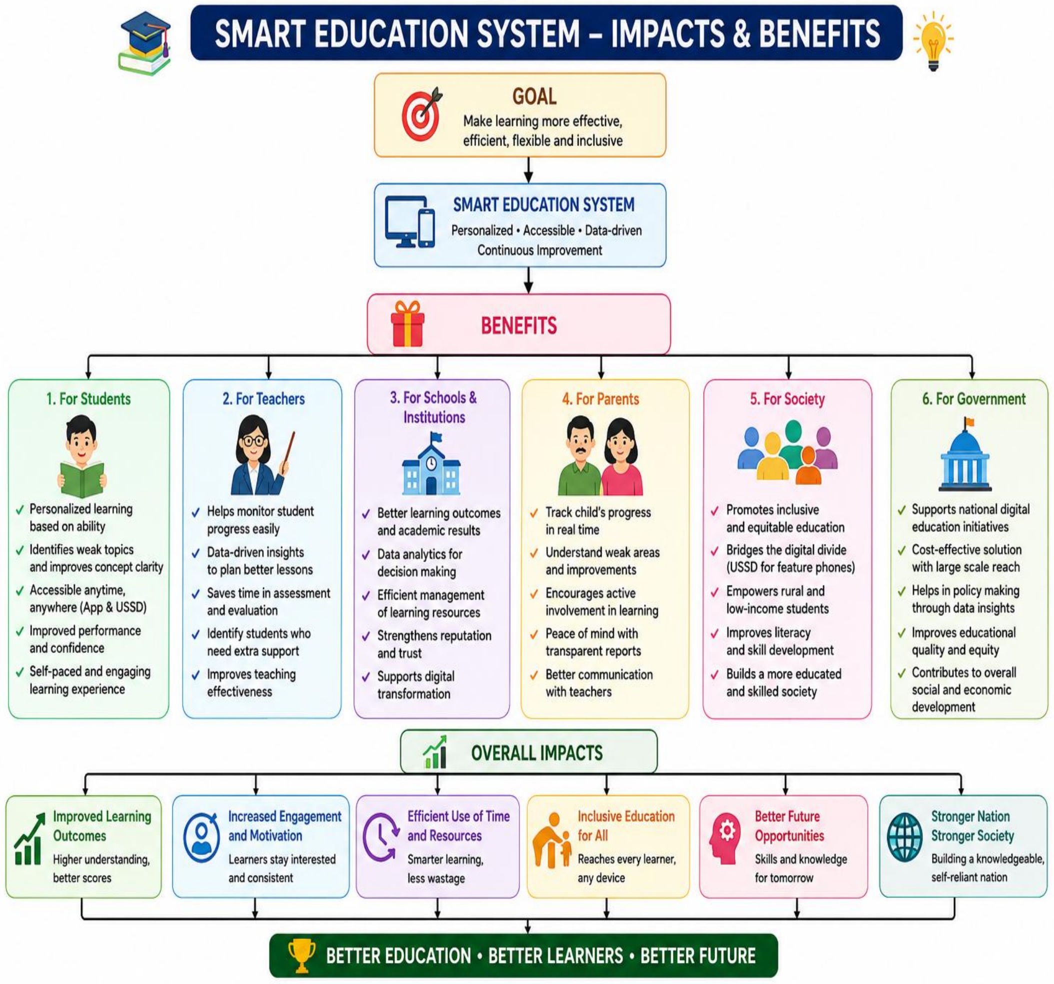
Potential Impact on Target Audience:

- **Rural & Underprivileged Students:** Brings self-paced diagnostic learning to learners without smartphones or internet access via 2G USSD/SMS.
- **School & College Learners:** Eliminates exam stress by pinpointing exact weak concepts instead of forcing repetitive rote study.
- **Teachers & Educators:** Automates homework grading and provides class-level analytics on common conceptual stumbling blocks.

Key Benefits of the Solution:

- **Social Equity:** Bridges the digital divide by making quality educational tools available to any basic mobile phone.
- **Time Efficiency:** Reduces student study time by focusing exclusively on unmastered prerequisites and weak topics.
- **Cost Effective:** Built entirely with free, open-source tools with zero data cost for 2G USSD users.



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RESEARCH AND REFERENCES

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POLICY & EDUCATIONAL FRAMEWORKS

- **National Education Policy (NEP 2020):** Aligned with core recommendations for adaptive technology, formative assessment, and digital equity across rural and urban institutions.
- **Bloom's 2 Sigma Problem & Mastery Learning:** Grounded in research showing that 1-on-1 personalized diagnostic tutoring and immediate feedback dramatically improve student test performance.

TELECOMMUNICATION & ACCESSIBILITY STANDARDS

- **3GPP Technical Specifications for USSD (TS 22.090 / TS 23.090):** Standards governing real-time, low-latency, session-oriented text communication over 2G/3G cellular signaling channels.
- **W3C Web Content Accessibility Guidelines (WCAG 2.1):** Standards used to ensure responsive layouts, high contrast, and voice-assisted navigation for all student interfaces.

TECHNICAL DOCUMENTATION

- **Python Software Foundation & Flask Framework Documentation** (Backend API development & session routing)
- **Scikit-Learn Machine Learning Documentation** (Classification models & adaptive performance analysis)
- **SQLite / PostgreSQL Official Reference Manuals** (Relational schema & offline-first data synchronization)

KEY INNOVATIONS

- **Root-Cause Detection:** Isolates basic formula gaps before teaching advanced concepts.
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